

Title:**Design & Development of an Ultra Low-Cost Active Miniature RFID Sticker-Type Self-Powered Anti-Theft and Asset Management Device for Hotel Items****IOT Number: IOT****Project Description:**

This project proposes the design and development of an ultra low-cost smart RFID-based anti-theft and asset management system intended for hotel room items such as electric kettles and TV remotes. The system is designed to detect unauthorised removal of hotel assets, trigger immediate audible alarms, and automatically log every event to a cloud-based Google Sheets database with accurate IST timestamps.

In many hotels, small assets are frequently misplaced or stolen without any tracking mechanism. Conventional RFID systems are often expensive or rely on proprietary middleware software. To overcome this issue, the proposed system integrates passive RF energy harvesting, IoT-based cloud logging, and real-time bidirectional status tracking into one compact and affordable solution.

The system is built around an ESP32 microcontroller, which acts as the central controller. It continuously reads data from the RC522 RFID reader, performs intelligent asset identification, triggers alarms, and sends structured log data to Google Sheets via WiFi.

Main Components Used:

- ESP32 Dev Board – Main controller, WiFi communication, and processing unit
- RFID RC522 Module – 13.56 MHz MIFARE reader for passive tag detection
- Blue RFID Keyfob Tag – Passive tag attached to Item A (Electric Kettle)
- White RFID Card – Passive tag attached to Item B (TV Remote)
- Active Buzzer – Provides audible alarm on unauthorised removal detection
- LCD 16×2 + I2C Module – Displays live asset status on-device
- Google Sheets + Apps Script – Free cloud database for real-time IoT logging

Working Principle:

The RC522 RFID reader continuously polls for tags within its 0–5 cm RF field. When a registered tag is detected, the ESP32 maps the UID to its product name, product ID, and room number from an internal lookup table. The system then fetches the current IST date and time from an NTP server via WiFi, determines the asset status, and sends a structured HTTP POST request to Google Apps Script, which appends a colour-coded row to the live Google Sheet.

Under Normal Conditions (Item Recovered):

- LCD displays item name and 'Recovered OK'
- Two short buzzer beeps confirm successful scan
- Google Sheet logs: Product Name | Product ID | Room No | Date | Time | Recovered

Under Fault Conditions (Item Missing / Theft):

If a registered tag is removed from the reader zone without authorisation:

- Buzzer sounds 4-beep alarm sequence for immediate audible alert to hotel staff
- LCD displays item name and 'MISSING! ALERT!'
- Google Sheet logs the event with red-highlighted status: Missing
- Complete timestamp record is preserved for audit and insurance purposes

Main Objective:

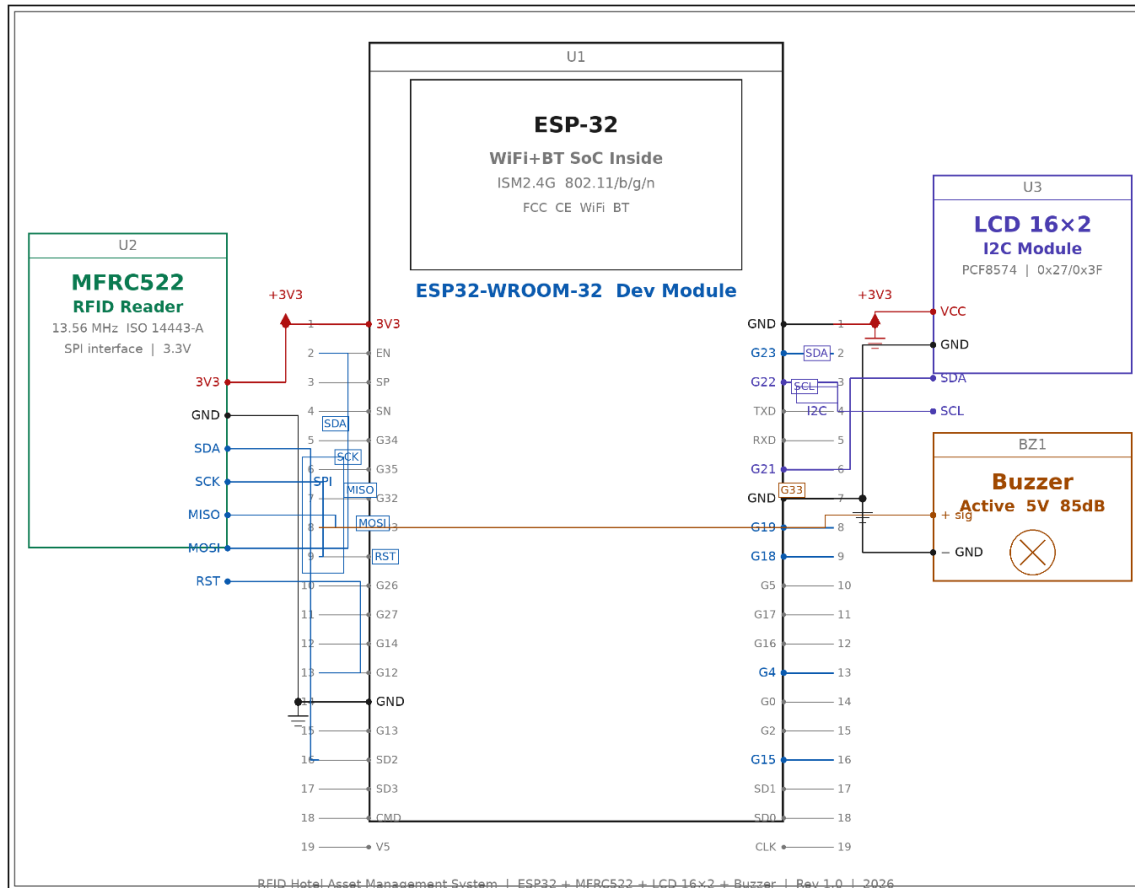
To provide a cost-effective intelligent RFID IoT asset protection system for hotels that combines:

- Passive RF energy harvesting — zero per-tag battery cost or maintenance
- Real-time bidirectional status tracking — both Missing and Recovered events logged
- NTP-synchronised IST timestamps on every cloud log entry
- Zero middleware cost — Google Sheets replaces expensive enterprise RFID software
- Edge security — UIDs resolved locally on ESP32, only human-readable data sent to cloud
- End-to-end latency under 3 seconds from tag scan to Google Sheets row

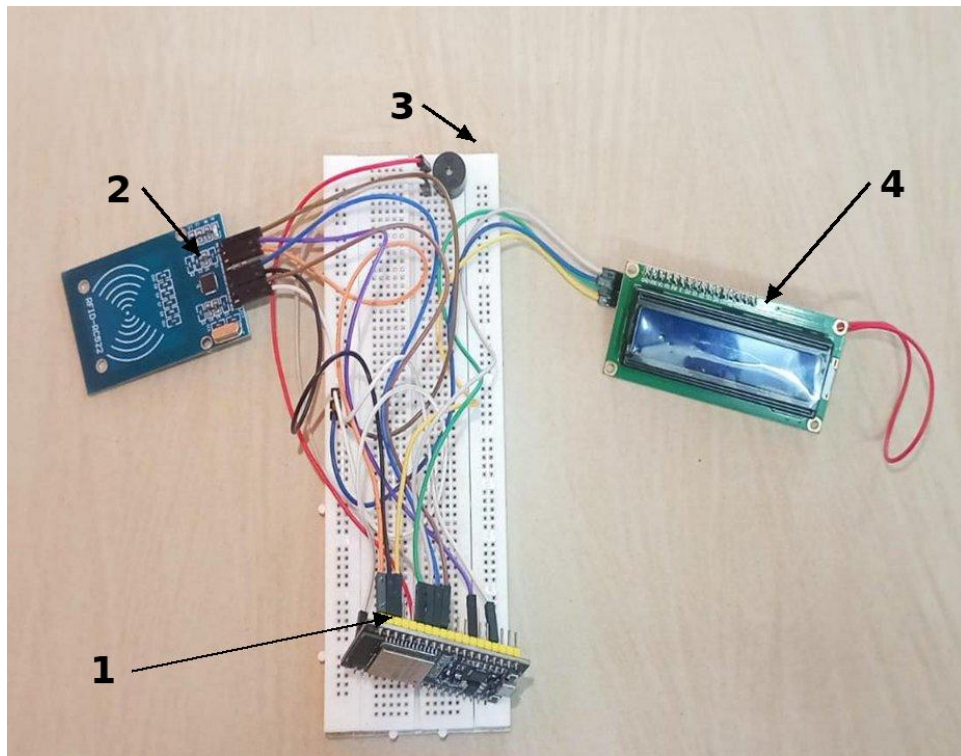
Applications:

- Hotels and resorts — protection of room assets (kettles, remotes, hairdryers)
- Hospitals — medical equipment tracking and theft prevention
- Libraries — book and resource asset management
- Small offices and co-working spaces — inventory audit systems
- Hostels and dormitories — shared item management
- Retail stores — anti-theft and stock management

Circuit Diagram:



Prototype:



Components:

S.No	Components
1	ESP32 WiFi Module
2	RFID RC522 Reader Module
3	Active Buzzer
4	LCD 16×2 Display + I2C Module

Component Descriptions:

1. ESP32 Dev Board:

The ESP32 is the main control unit of the system. It reads tag UID data from the RC522 RFID module via SPI, performs UID-to-product mapping locally, fetches real-time IST timestamps from an NTP server via WiFi (802.11 b/g/n), and sends structured asset log data to Google Sheets via HTTP POST through Google Apps Script. It also controls the buzzer alarm and LCD display. With dual-core 240MHz processing and built-in WiFi and Bluetooth 4.2, it provides a powerful and compact IoT platform for the entire system.

2. RFID RC522 Module:

The MFRC522 is a 13.56 MHz RFID reader module operating on the ISO 14443-A protocol, compatible with MIFARE Classic 1K passive tags. It communicates with the ESP32 via the SPI bus on GPIO pins 15 (SDA/SS), 18 (SCK), 19 (MISO), 23 (MOSI), and 4 (RST). The module generates a continuous RF field that energises passive tags within 0–5 cm without requiring any battery on the tag itself, making it ideal for a maintenance-free deployment on hotel assets.

3. Active Buzzer:

The active buzzer provides audible alerts for both theft detection and successful recovery events. Connected to GPIO 33 of the ESP32, it emits a 4-beep alarm sequence when an item goes missing, and two short confirmation beeps when an item is recovered. At 85dB, it is loud enough to alert hotel staff across a room. Being an active type buzzer, it requires only a digital HIGH signal to operate without any additional PWM or tone generation circuitry.

4. LCD 16×2 Display + I2C Module:

The LCD 16×2 display shows real-time asset status directly on the device without requiring a laptop, phone, or separate monitoring station. The I2C backpack module (PCF8574) reduces the wiring from 8–16 parallel data lines to just 4 wires: VCC, GND, SDA (GPIO 21), and SCL (GPIO 22), minimising GPIO usage on the ESP32. It displays startup messages, WiFi connection status, item names, and alarm states with every scan. The I2C address is configurable between 0x27 and 0x3F depending on the hardware variant.

OUTPUT:

The Google Sheets dashboard provides a complete, colour-coded, real-time audit trail of all asset events. Every RFID scan automatically appends a new row with the product name, product ID, room number, date, time, and status — with the status cell highlighted red for Missing and green for Recovered. The screenshot below shows actual logged data from the system running live.

ASSET MANAGEMENT															
File Edit View Insert Format Data Tools Extensions Help															
100% \$ % .00 123 Default... - 10 + B I A															
A1	DATE														
21	27/04/2026	16:12:13	P001	Electric Kettle	Room 204	Missing									
22	27/04/2026	16:12:27	P002	TV Remote	Room 204	Missing									
23	27/04/2026	16:12:42	P002	TV Remote	Room 204	Recovered									
24	27/04/2026	16:12:51	P001	Electric Kettle	Room 204	Recovered									
25	27/04/2026	16:13:20	P001	Electric Kettle	Room 204	Missing									
26	27/04/2026	16:13:39	P002	TV Remote	Room 204	Missing									
27	28/04/2026	15:56:34	P001	Electric Kettle	Room 204	Recovered									
28	28/04/2026	15:56:53	P002	TV Remote	Room 204	Recovered									
29	28/04/2026	15:57:05	P001	Electric Kettle	Room 204	Missing									
30	28/04/2026	15:57:19	P002	TV Remote	Room 204	Missing									
31	28/04/2026	15:57:41	P001	Electric Kettle	Room 204	Recovered									
32	28/04/2026	15:57:53	P002	TV Remote	Room 204	Recovered									
33	08/05/2026	10:08:37	P002	TV Remote	Room 204	Recovered									
34	08/05/2026	10:11:19	P002	TV Remote	Room 204	Missing									
35	08/05/2026	10:11:59	P001	Electric Kettle	Room 204	Recovered									
36	08/05/2026	10:13:56	P002	TV Remote	Room 204	Recovered									
37	--/--/--	--/--/--	P002	TV Remote	Room 204	Recovered									
38	--/--/--	--/--/--	P002	TV Remote	Room 204	Missing									
39	--/--/--	--/--/--	P002	TV Remote	Room 204	Recovered									
40	--/--/--	--/--/--	P001	Electric Kettle	Room 204	Recovered									
41															
42															
43															

As shown above, the system successfully logged multiple Missing and Recovered events for both the Electric Kettle (P001) and TV Remote (P002) across multiple dates (27/04/2026, 28/04/2026, 08/05/2026) with accurate IST timestamps. The colour-coded status cells — red for Missing and green for Recovered — create a clear, readable audit trail.